

100 Days of Linux - Day 043

Title: Creating and Extracting ZIP Archives

Today's Goal

Learn how to create, update, list, and extract ZIP archives using the **zip** and **unzip** commands.

Why This Matters

ZIP is the most common archive format for sharing files between Windows, macOS, and Linux. Developers and data engineers often use it to exchange projects and datasets.

Commands of the Day: zip & unzip

Common Syntax:

```
zip archive.zip file1 file2
zip -r project.zip project/
unzip archive.zip
unzip -l archive.zip
zip -u archive.zip newfile.txt
```

Options:

- r = Recursively include folders
- l = List archive contents
- u = Update an existing ZIP archive

Beginner Lesson

Unlike tar, ZIP combines archiving and compression in one format. It is widely supported across operating systems and is commonly used for sharing projects.

Practice Questions

1. Create a folder named `project_files` containing `README.md`, `employees.csv`, and `notes.txt`. Compress it into `project_files.zip`.
2. List the contents of `project_files.zip` without extracting it.
3. Extract `project_files.zip` into a folder named `extracted_project`.
4. Create a file named `changelog.txt` and add it to the existing ZIP archive using the update option.
5. Display your current working directory and list all ZIP files.

Hints

Use `zip -r` for folders, `unzip -l` to inspect archives, and `zip -u` to update an existing archive.

Mini Challenge

Create a `retail_backup` folder with data, logs, and scripts subfolders. Compress it into `retail_backup.zip`, extract it into `restore/`, then verify the files with `ls`.

Common Mistakes

Forgetting the -r option when compressing directories, which results in an incomplete archive.

Did You Know?

Many organizations exchange datasets and reports as ZIP files because almost every operating system supports them.

Pro Tip

Before sending a ZIP archive, use `unzip -l` to verify that it contains all the expected files.

Answer Key (Instructor Only)

Q1: `zip -r project_files.zip project_files/`

Q2: `unzip -l project_files.zip`

Q3: `mkdir extracted_project && unzip project_files.zip -d extracted_project/`

Q4: `zip -u project_files.zip changelog.txt`

Q5: `pwd && ls *.zip`

Homework

Create three different project folders and practice creating, listing, updating, and extracting ZIP archives.

Next Day Preview

Tomorrow you'll complete a practical compression and archiving assignment using `tar`, `gzip`, `gunzip`, `zip`, and `unzip`.